

5	(a)	The first order maximum is diffracted at an angle $\theta = \tan^{-1} (0.46/2.00) = 12.95^\circ$. From $d \sin \theta = n\lambda$ where $d = 1/N = 1/300 \times 10^3$ and $n = 1$ $\frac{\sin 12.95}{300 \times 10^3} = 1 \times \lambda$ $\Rightarrow \lambda = 7.47 \times 10^{-7} \text{ m}$	
	(b)	Using the second-order diffracted light to measure the wavelength is more accurate. This is because the larger angle of diffraction can be measured experimentally with a lower percentage error for a given precision of the measuring instrument used.	

6	(a)	Use the formulae in order to calculate the length of the slit.	
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